

Karan Sidhu

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Education

University of Ottawa

Master of Computer Science

Sep. 2026 – Expected 2028

Concentration in **Applied Artificial Intelligence**

University of Alberta

Bachelor of Science | GPA: 3.3

Sept 2020 - Jun 2025

Double Major in **Computer Science and Biology**

Relevant Coursework: search algorithms, supervised learning, unsupervised learning, reinforcement learning, parsing/tagging texts, error analysis, text normalization, chatbots, storage architecture design, UML, software design patterns, software architecture, object-oriented programming, Design Patterns, Storage Architecture, Natural Language Processing.

Skills

Programing: Python, Java, ReactJS, Angular, html5, CSS3, C

DBSM: SQL, MongoDB, Firebase, Neo4j Cypher (GraphQL)

AI/ML: Prompt engineering, fine-tuning ML models, relation extraction, chatbots

Software: git, GitHub, bitbucket, Huggingface, Pytorch, Tensorflow, linux command line and bash scripting

Design: Figma, UI/UX design, wireframing, user flow design, Responsive Design

Experience

Public Health Agency of Canada (PHAC) | Website Developer

Sept 2026 - Present

- Developing a survey table system.

RoboGarden | UI/UX Developer

Jul 2026 - Aug 2026

- Authored technical review reports for 10 online courses (8 CodingVille, 2 UpSkilling), evaluating course design, quiz UX, and content structure.
- Applied UI/UX principles while building proficiency in Angular and modern full-stack web development.
- Designed high-fidelity Figma prototypes for an interactive table-editing interface with a focus on usability and accessibility.
- Developed a pagination table application with full CRUD functionality.
- Conducted competitive UX research to inform product design decisions.
- Created user flows and wireframes in Figma for the RoadMap project, improving navigation and learning progression.
- Designed and integrated new Mentor and Messaging features, allowing students to request mentorship from course pages and enabling mentors to manage roadmaps, assign lessons, leave direct comments, and communicate via a real-time messaging system.
- Collaborated with stakeholders to refine requirements and finalize designs, while supporting front-end developers to ensure design consistency.

BlessRhooDayCare | Website/Business Developer ([URL](#))

Apr 2026 - Jun 2026

- Designed and deployed a responsive business website using React, TypeScript, and Tailwind CSS to modernize the brand's digital identity.
- Integrated interactive Google Maps within the site to enhance local visibility, simplify location finding for prospective clients, and improve user engagement.
- Engineered a seamless, backend-free lead generation pipeline by integrating custom form validation and email handling via Formspree.
- Increased local discoverability by architecting and launching the business's Google Maps/Search footprint through a fully optimized Google Business Profile.
- Streamlined business operations by implementing professional email infrastructure and providing end-user training on modern document management workflows.

- Managed the end-to-end technical lifecycle, from requirements gathering with stakeholders to production deployment on modern hosting platforms.

TMIC Wishart Node | Artificial Intelligence/Software Developer May 2024 - Sep 2024

- Delivered weekly presentation project updates, and ensured alignment on research goals.
- Enhanced knowledge graph construction through an AI pipeline that extracts data (triplets) from research articles using LLM's.
- Fine-tuned HuggingFace models, improving relation extraction accuracy by 14.6%.
- Engineered a cost-effective alternative to AppSignal for error and performance monitoring.
- Designed custom GPT chatbots to streamline and automate the grant application processes.

ACE Liquor Discounter | Team Member Apr. 2021 - Aug. 2023

- Independently opened and closed the store as a trusted keyholder, overseeing daily operations.
- Managed cash, processed transactions, and completed end-of-day reconciliation.
- Ordered inventory, coordinated deliveries, and stocked products to maintain availability and freshness.
- Delivered excellent customer service, professionally handling difficult and intoxicated customers.
- Responded to theft incidents and supported loss prevention procedures.
- Maintained store cleanliness and compliance with liquor control and safety regulations.

Projects

ChemFont Fact Finder | ([Fact Finder](#), [Project7](#), [TextMining Chatbot](#)) Jan 2024 - Sep 2024

- Designed and implemented a pipeline to extract triplets from research articles.
- Developed high-quality datasets that enhanced evaluation accuracy and model assessments.
- Engineered a system of bonus positives to refine evaluation metrics, addressing previously unaccounted valid predictions and improving model assessment reliability.
- Automated quantitative evaluation, increasing accuracy and consistency of model assessments.
- Actively worked with OpenSource Models (LLaMA, BERT, Mistral, Mixtral, ect.)

AppSignalReplicate | ([App Signal Replicate](#)) Jul 2024 - Aug 2024

- Engineered a lightweight, free alternative to AppSignal for error tracking and performance monitoring across multi-server environments.
- Automated log analysis across servers via SSH with real-time email notifications for anomalies, ensuring proactive error detection.

Event QR Check-in App | ([QR Check-in App](#)) Jan 2024 - Jun 2024

- Designed and developed an Android app for event management using Android Studio and Firebase, enhancing event check-ins for attendees, organizers, and admins.
- Engineered QR code generation and scanning modules, integrating them with real-time database queries to support event workflows.
- Oversaw agile cycles and tracked tasks with GitHub Projects, ensuring delivery of milestones.

Portfolio Chatbot Website | ([Public Portfolio](#)) May 2025 - Jun 2025

- Built a modern, responsive AI chatbot website using React 18, Vite, and Tailwind CSS for fast performance and a seamless user experience.
- Integrated dynamic routing with React Router and modular architecture with reusable components for scalability.
- Deployed backend chatbot service alongside a styled frontend based on community Figma designs, enabling user-friendly conversational AI access.